Appendix to "Representable but not found: what path-integrated rotary attention learns, and how it fails"

Draft 1 (2026-10-18), for `docs/paper/MAIN_DRAFT1.md`. The full theory appendix, with every check script and the results not used here (P1(c)-(e), P2, P5, P7, Remark R), is `docs/paper/APPENDIX_theory_v3.md`; App. E below states and proves what the main text uses and gives the v3 label of each statement. Every number is copied from the file named beside it; a CORRECTED or AUDIT block at the top of a results file supersedes its body. "Registered" = a pre-registered decision or declared secondary; "(analysis)" = computed after the batch on stored streams or checkpoints, never changing a verdict. Delete this paragraph before submission.

Contents. A: models, recipes, batches. B: floors. C: per-seed tables, SOLVED-threshold sensitivity, evaluation scale. D: constructions. E: theory. F: exchange-rate details. G: rank details. H: sign details. I: CARoPE and CoPE. J: text world. K: what and where. L: PoPE's score.

A. Models, recipes, batches

A1. Architecture

component	setting
layers	1 (Sec. 5: 1-4 real layers and one block looped x4; ROBUST torus: 2 layers, d_model 256)
heads, width	2 heads, d_model 128, d_head 64; feed-forward 4 x d_model with GELU; LayerNorm before attention and before the feed-forward; dropout 0.1 on attention weights, attention output and feed-forward output
rotary channels	32 two-dimensional blocks per head, 64 channels; every query and key coordinate is rotated
score	channel c contributes q_c^T R(theta_{t',c} - theta_{t,c}) k_c (content Q and K rotated by the path angle; not additive)
step map	Delta = W_out W_in x, W_in: 128 -> r, W_out: r -> 64, no bias; r = 4 shared across heads unless stated (MapFormer's latent is per head); per-head variants: one W_in / W_out block per head; block-diagonal rank 4: the shared rank-4 latent with the off-diagonal blocks of W_out held at zero (each head reads rank 2)
frequencies	omega_c = 2 pi (1 / Delta_max)^(i / 31) for block i = 0..31, the same ladder in each head; Delta_max = grid size (torus 64, ring 32, line 256); omega is a free parameter, its sign unconstrained
vocabulary	torus: 4 moves + 17 observation tokens (16 types + blank); ring and line: 2 moves + 17
parameters	about 204k for the one-layer torus MapWM; 798,741 for four index layers; 203,925 for the looped block (`FACTORIAL_RESULTS.md`); 1.59M for the ROBUST torus model

Step parameterisations (`model.py`, `model_sign.py`, `model_wrap_exist.py`): signed (`Signed_r4` = `Vanilla_r4`), |.| (`Abs_r4`; the absolute value is applied after W_out on each of the 64 channel steps, adding no parameters), softplus (`Pos_r4`; softplus(0) = 0.693, so every token starts at about the same step), observation-step-free (`SignedAO_r4`, `AbsAO_r4` on the ring,

`SignedAO4_r4`, `AbsAO4_r4` on the torus: every non-move token's step is 0 by construction). Index RoPE (`RoPE`): canonical ladder base^(-2c / d_head), base 10000.

A2. Recipes

	torus (Secs. 4-6, 7)	**ring and line (Secs. 5, 6)**	**Dyck-2 (Sec. 5)**
optimiser	AdamW, weight decay 0.05	AdamW, weight decay 0.05	AdamW, weight decay 0.01
peak learning rate	1e-3	1e-3 (ROBUST and LRSWEEP vary it)	1e-4
schedule	5% linear warm-up, cosine to 10% of the peak	same	same
batch	16 walks of T = 1024 moves (2,048 tokens); T = 128 batches of 128	128 walks of T = 128	128 sequences
budget	98 batches per epoch, 900 epochs (1800 in Sec. 6.1; 300 at T = 128)	98 batches per epoch, 300 epochs	560,000 sequences (1x) or 1,680,000 (3x)
loss	cross-entropy on revisited observations	same	next-token cross-entropy on every position
training map	one per run	redrawn for every walk	--
evaluation	100 walks on one held-out map, eval mode	400 walks, maps redrawn	L32 D12 test set

Run classes (`stats_core.classify_run`, per-epoch training loss): SOLVED if the mean of the final 5% of epochs is below 0.05; otherwise STALLED if the final-10% mean is within 5% of the preceding 10% mean, else DESCENDING. Continuations of constructed models (App. D): ring 300 epochs at lr 1e-3 (schedule re-warmed, fresh optimiser state, fresh walk stream); torus 300 epochs at lr 1e-4, fixed before launch (about the rate at which the source solution was reached).

A3. Batches

"Code check" = an independent, results-blind code review, whose findings were amended into the pre-registration before any result was read (before launch, except TEXTWORLD and LEAK, checked while the batch trained): a separately launched read-only AI code-review session given the pre-registration and the code but no results. "Reproduced" = the analysis was re-run from stored checkpoints by an independent reproduction audit (`docs/paper/review_draft5/REPRO_AUDIT.md`, `REPRO_AUDIT_nonsign.md`, 2026-10-16) and matched byte for byte (spot re-evaluations within 7e-4). "Attention": standard, or `--fast-attn` (SDPA; agrees to ~1e-6 in logits, not bit-identical in training). Pilots ran on seeds outside the batch except where noted.

batch	**main text**	**task**	**arms**	**seeds**	**epochs**	**attention**	**code check**	**reproduced**
PAPER2X2	4	torus T = 128	RoPE, PoPE, MapWM r2 / r4, MapPoPE r2 / r4	0-7	300	standard	no	yes

batch	main text	task	arms	seeds	epochs	attention	code check	reproduced
SIGN_MATCHED	4, 6.2	torus T = 1024	signed, |.|, softplus, RoPE	0-7	900	standard	no (authors' pre-launch gate)	yes
GENERALITY	4, 6.2	torus; ring p_plus 0.5	obs.-step-free signed / |.| and constructions; signed, CARoPE, CoPE, RoPE	700-707	900 (ring 300); +300	fast-attn (torus)	yes	yes
FACTORIAL	5	torus T = 1024	path 1 layer; index 1, 4 layers, looped x4	500-503 (n = 4)	900	fast-attn	yes	yes
DYCK_MDEPTH	5	Dyck-2 L32 D12	MapWM, MapPoPE (r2), RoPE, PoPE at 1-4 layers; 3x; mixture	0-7	1x / 3x	standard	no	yes
CANCEL	5	ring p_plus 0.5-1.0	path 1 layer; index 1-3 layers	0-7; main text 2-7 (0-1 were the pilot)	300	standard	no	yes
RANK_MI	6.1	torus	shared r2, per-head r2, shared r4	0-7 (per-head r2 s0-1 were a pilot)	900	standard	no	yes
RANK_SEP	6.1	torus	block-diagonal r4, per-head r4	0-7	900	standard	no	yes
RANK3	6.1	torus	per-head r3	0-7	900	standard	no	yes
LOOP_RANK_E1800 (part 1)	6.1	torus	shared r2, shared r4	0-7	1800	standard	no	yes
RANK_PROJ	6.1	torus	projected r2 (frozen, trained), r4 control, r2 continued	0-7	+900	standard	no	yes
RANK_ND	6.1	2D (32 per side), 3D (10 per side) tori	per-head rank D, D + 1	0-7	900	standard	yes	yes
RANK_WRAP	6.1	2D and 3D tori	per-head rank D + 1	0-7	900	standard	yes	yes

batch	main text	task	arms	seeds	epochs	attention	code check	reproduced
		of two sizes						
RANK_TESTS	6.1	ring; torus escape test	rank 1, 2 (+ variants); lifted / fresh rank 4	20-27; 0-7	300; 600	standard	yes	yes
RING_DIAL	6.2	ring p_plus 0.5 / 0.75 / 0.9	signed, |.|	30-37	300	standard	yes	yes
WRAP_EXISTENCE	6.2	ring p_plus 0.5	obs.-step-free signed / |.|, constructions	600-607	300 + 300	standard	yes	yes
LRSWEEP	6.2	ring	obs.-step-free signed / |.| at 1e-3 / 3e-3 / 1e-2 / 3e-2	1500-1507	300	standard	yes	no (windowed run classes cross-checked)
ROBUST	6.2	ring (four recipes); torus 2 layers d 256	obs.-step-free signed / |.|	1400-1407	300; 900	fast-attn (torus)	yes	yes
ACCOUNTS	6.2	ring; line	T1 continuations; T2 frozen latents; T4 CoPE on a line	600-607; 1000-1007; 1100-1115	300	standard	yes	yes
MECHANISM	6.2	ring p_plus 0.5	eight start / step arms	800-807	300	standard	yes	yes
TEXTWORLD	7	torus in words	path 1 layer; index 1, 2 layers	0-7; main text 2-7 (0-1 were the pilot)	900	standard	yes	yes
RING_BUDGET (H)	--	ring p_plus 0.5	|.| at 300 and 900 epochs	40-47	300 / 900	standard	yes	yes
TORUS_START (H)	--	torus	start arms	1200-1207	900	fast-attn	yes	yes
GAUGE (H)	--	ring, moves +1/-1 or +1/+2	obs.-step-free signed / |.|, RoPE	1300-1307	300	standard	yes	yes
WRAP_COPE (I)	--	ring	CoPE table starts	900-907	300	standard	yes	yes
LEAK (K)	--			0-7	900	standard	yes	yes

batch	main text	task	arms	seeds	epochs	attention	code check	reproduced
		new-object torus	MapWM, ActOnly, NormStep					
GAIN_PHASE (K)	--	new-object torus	step x score 2x2	8-15	900	standard	yes	no
MAPPOPE_PAIR (L)	--	torus T = 128	MapWM r2, MapPoPE r2 (32 / 64 angles)	10-25	300	standard	yes	no

ACCOUNTS T1's weight-decay-0.05 arms are bit-identical retrains of WRAP_EXISTENCE's constructed arms (a second readout of the same runs, not a replication). The shared rank-2 arm of every rank contrast is one stored set of eight runs.

A4. Reproduction of MapFormer v4 Table 2 (2D)

With the paper's recipe (16 epochs x 98 batches, linear decay, lr 3e-4; `python3 -m mapformer.main --device cuda --epochs 16 --n-batches 98`), our MapWM scores 0.989 +/- 0.010 and our MapEM (single p_0) 0.987 +/- 0.012 on fresh maps at T = 128 (`PAPER_TASK_ACCURACY.md`), against 1.00 / 1.00 in v4 and 0.99 / 1.0 in v1 (MapWM-r2, MapEM-os, 2D IID).

B. Floors

A floor is the accuracy of the best map-free predictor on the same kind of evaluation walk. Rows marked (analysis) were computed after the batches.

task	predictor	value	stream	source
torus T = 128 (analysis)	blank / n-gram (orders 0-5) / retrace	0.507 / 0.598 / 0.934 (retrace per seed 0.928-0.946); wrap-only share 0.000	PAPER2X2 eval walks, seeds 0-7	`docs/audits/2026-10-16/paper2x2_T128_floors_out.txt`
torus T = 1024 (analysis)	blank / n-gram / retrace	0.506 / 0.575 / 0.843 (retrace per seed 0.837-0.849); wrap-only share 0.079	SIGN_MATCHED eval walks, seeds 0-7 (shared by RANK_MI, RANK_SEP, RANK3)	`docs/audits/2026-10-16/sign_floors_out.txt`
torus T = 1024 (analysis)	same	0.508 / 0.578 / 0.843; wrap-only 0.081	GENERALITY, seeds 700-707	`generality_floors_out.txt` (same directory)
torus T = 1024, 2 layers d 256 (analysis)	same	0.509 / 0.578 / 0.844; every run of both arms above retrace (lowest |.| 0.910)	ROBUST, seeds 1400-1407	`robust_floors_out.txt`

task	predictor	value	stream	source
torus T = 1024 (analysis)	blank / retrace	0.509 / 0.844	FACTORIAL, seeds 500-503	`docs/paper/figs/fig_factorial_strata_out.txt`
ring p_plus 0.5 / 0.75 / 0.9 (analysis)	retrace	0.860 / 0.758 / 0.612 (blank 0.497-0.499)	2000 walks per bias	`docs/audits/2026-10-16/ring_floors_out.txt`
ring p_plus 0.5 / 0.75 / 0.9 / 1.0	better of best constant and n-gram (orders 1-5)	0.554 / 0.544 / 0.520 / 0.498	1000 held-out walks	`CANCEL_PREREG.md`, `CANCEL_RESULTS.md`
line (256 cells)	blank	0.503	ACCOUNTS T4	`ACCOUNTS_RESULTS.md`
ring, moves +1/+2	blank (retrace reduces to it)	~0.50	GAUGE	`GAUGE_RESULTS.md`
torus in words	best constant / reversal-copy	0.505 / 0.597	TEXTWORLD eval set	`TEXTWORLD_RESULTS.md`
Dyck-2 L32 D12	best stack-free A2f floor (n-grams of orders 1-6)	0.594	gate G4	`DYCK_MDEPTH_RESULTS.md`

Per-stratum floors (analysis). Torus, seeds 0-7 (`docs/paper/figs/torus_strata_floors_s0-7.json`): best constant 0.506 / 0.500 / 0.507 (short gap / long gap / wrap-only), retrace 0.911 on short gaps and equal to the constant on the others (walk runs are at most 10 moves, so a retrace never reaches a gap of 128 moves or a wrap-only revisit). FACTORIAL's stream: 0.508 / 0.515 / 0.517, retrace 0.912. Ring (`ring_strata_floors_out.txt`): retrace 0.9998 at gaps 2-7, 0.943 at 8-31, 0.570 at 32-127, 0.493 wrap-only; constant 0.49-0.50 in every stratum.

Against its own seed's retrace floor (analysis, torus T = 1024): signed falls below on 0/8 runs, |.| on 3/8 (0.637, 0.690, 0.717), softplus on 4/8, index RoPE on 8/8.

C. Per-seed tables

Accuracy at the training length on held-out maps, eval mode. SOLV / STAL / DESC = SOLVED / STALLED / DESCENDING; a number in brackets is the final-5% training loss.

C1. Torus sign batch (Secs. 4, 6.2; `SIGN_MATCHED_ANALYSIS.txt`)

seed	signed	|.|	softplus	index RoPE
0	1.000 SOLV (0.003)	0.931 DESC (0.201)	0.590 STAL (1.250)	0.730 STAL (0.983)
1	0.997 SOLV (0.003)	0.917 DESC (0.314)	0.527 STAL (1.510)	0.729 STAL (0.939)

seed	signed	|.|	softplus	index RoPE
2	1.000 SOLV (0.005)	0.868 STAL (0.492)	0.630 STAL (1.084)	0.720 STAL (0.931)
3	1.000 SOLV (0.008)	0.637 STAL (1.043)	0.972 DESC (0.158)	0.739 STAL (0.910)
4	1.000 SOLV (0.004)	0.690 DESC (0.962)	0.674 STAL (1.009)	0.737 STAL (0.928)
5	0.990 SOLV (0.003)	0.888 STAL (0.446)	0.935 DESC (0.295)	0.744 STAL (0.885)
6	1.000 SOLV (0.004)	0.922 STAL (0.349)	0.922 DESC (0.345)	0.722 STAL (0.956)
7	1.000 SOLV (0.004)	0.717 STAL (0.834)	0.995 SOLV (0.049)	0.728 STAL (0.976)
mean	0.998	0.821	0.781	0.731

Registered: |.| - signed -0.177 (permutation p 0.0002; 0/8 vs 8/8, Fisher p 0.0002); softplus - signed -0.218 (p 0.0003; 1/8 vs 8/8, p 0.0014); signed - RoPE +0.267 (p 0.0002; 8/8 vs 0/8, p 0.0002). r(final loss, accuracy) -0.992 over 32 runs.

C2. Torus, index depth (Sec. 5; `FACTORIAL_{L1,L4,LOOP}.json`)

seed	path, 1 layer	index, 1 layer	index, 4 layers	index, looped x4
500	1.000	0.719	0.814	0.868
501	1.000	0.731	0.870	0.861
502	1.000	0.688	0.874	0.860
503	1.000	0.739	0.881	0.868

Every path run SOLVED, every index run STALLED. Registered C0-C3: path beats index at one layer (+0.281), four layers (+0.141) and loop x4 (+0.136); depth and loop each help index (+0.140, +0.145); every firing a complete separation (p 0.029).

C3. Ring exchange rate, fresh seeds 2-7 (Sec. 5; `docs/audits/2026-09-27/fresh_seeds.txt`)

p_plus	path 1 layer	index 1 layer	gap to path at 1 / 2 / 3 index layers
0.5	1.000	0.710 +/- 0.020	+0.290 / +0.051 / +0.002
0.75	1.000	0.677 +/- 0.009	+0.323 / +0.038 / +0.002
0.9	1.000	0.824 +/- 0.007	+0.176 / +0.013 / +0.001
1.0	1.000	1.000	0 / 0 / 0

All eight seeds (`CANCEL_RESULTS.md`): index 1 / 2 / 3 layers 0.717 / 0.954 / 0.998 (p_plus 0.5), 0.676 / 0.969 / 0.998 (0.75), 0.825 / 0.988 / 0.999 (0.9), 1.000 (1.0); path 1.000 in every cell (32/32 SOLVED); registered exchange rate k = 3, 3, 3, 1 index layers.

C4. Rank (Sec. 6.1; `RANK_MI_ANALYSIS.txt`, `RANK_SEP_ANALYSIS.txt`, `RANK3_ANALYSIS.txt`, `RANK_SEP.json`, `RANK3.json`)

seed	shared r2	per-head r2	block-diag r4	per-head r3	shared r4	per-head r4
0	0.849 STAL (0.587)	0.656 STAL (1.049)	0.879 STAL (0.482)	0.942 DESC (0.243)	1.000 SOLV (0.004)	0.996 SOLV (0.004)
1	0.998 DESC (0.051)	1.000 SOLV (0.007)	0.936 DESC (0.302)	0.990 SOLV (0.003)	1.000 SOLV (0.003)	1.000 SOLV (0.010)

seed	shared r2	per-head r2	block-diag r4	per-head r3	shared r4	per-head r4
2	0.917 DESC (0.308)	1.000 SOLV (0.003)	1.000 SOLV (0.010)	1.000 SOLV (0.003)	1.000 SOLV (0.003)	1.000 SOLV (0.004)
3	0.917 DESC (0.321)	0.723 STAL (0.910)	0.945 DESC (0.234)	0.967 DESC (0.134)	1.000 SOLV (0.003)	1.000 SOLV (0.004)
4	0.947 DESC (0.177)	0.959 DESC (0.194)	0.964 DESC (0.202)	1.000 SOLV (0.003)	1.000 SOLV (0.002)	0.998 SOLV (0.002)
5	0.907 STAL (0.404)	0.949 DESC (0.211)	0.917 DESC (0.310)	1.000 SOLV (0.006)	0.994 SOLV (0.004)	1.000 SOLV (0.008)
6	0.773 STAL (0.675)	0.943 DESC (0.248)	1.000 SOLV (0.010)	1.000 SOLV (0.003)	1.000 SOLV (0.004)	1.000 SOLV (0.005)
7	0.842 STAL (0.605)	0.851 STAL (0.539)	0.947 DESC (0.166)	1.000 SOLV (0.005)	0.987 SOLV (0.010)	1.000 SOLV (0.003)
mean	0.894, 0/8	0.885, 2/8	0.948, 2/8	0.987, 6/8	0.998, 8/8	0.999, 8/8

Per-head r2 seeds 0-1 were a pilot run before registration (1/6 SOLVED on seeds 2-7). Registered: per-head rank 4 vs block-diagonal 8/8 vs 2/8 (Fisher and permutation p 0.0070); shared r4 vs block-diagonal the same (it also changes per-head rank); sharing (per-head r4 vs shared r4, +0.001) and W_out scale (block-diagonal vs per-head r2, +0.063) not detected; per-head r3 vs r2 +0.102 (permutation p 0.027; SOLVED 6/8 vs 2/8, Fisher p 0.13); r4 vs r3 +0.012, not detected. Shared r4 vs shared r2: 8/8 vs 0/8 (Fisher p 0.0002), +0.104 (p 0.0005).

Twice the budget (`LOOP_RANK_E1800_P1_ANALYSIS.txt`, 1800 epochs from scratch): shared r2 0/8 (final-5% losses 0.489, 0.383, 0.390, 0.340, 0.141, 0.231, 0.305, 0.571; 3 STALLED, 5 DESCENDING), accuracy 0.908; shared r4 7/8 (s5 STALLED at 0.180), 0.994.

C5. Sign, ring and torus (Sec. 6.2)

Ring dial (`RING_DIAL_VERDICTS.json`; full models, S = SOLVED):

seed	0.5 signed	0.5 |.|	0.75 signed	0.75 |.|	0.9 signed	0.9 |.|
30	1.000 S	1.000 S	1.000 S	0.994 S	1.000 S	1.000 S
31	1.000 S	0.977	1.000 S	0.958	1.000 S	0.999 S
32	1.000 S	0.987	1.000 S	0.957	1.000 S	0.885
33	1.000 S	0.964	1.000 S	0.954	1.000 S	1.000 S
34	1.000 S	1.000 S	1.000 S	0.964	1.000 S	1.000 S
35	1.000 S	0.944	1.000 S	0.967	1.000 S	0.825
36	1.000 S	0.977	1.000 S	0.993	1.000 S	0.960
37	1.000 S	0.931	1.000 S	0.958	1.000 S	0.964

Registered: p_plus 0.5 -0.028 (permutation p 0.007), 2/8 vs 8/8 (Fisher p 0.0070); 0.75 -0.032 (p 0.0002), 1/8 vs 8/8 (p 0.0014); at 0.9 the contrast did not reach the registered threshold (-0.046, below the 0.05 materiality; 4/8 vs 8/8, p 0.077).

Observation-step-free ring from scratch and constructed (`WRAP_EXISTENCE_ANALYSIS.txt`):

seed	signed	|.|	constructed |.|: frozen -> trained (max continuation loss)	constructed signed: frozen -> trained
600	1.000 S	0.905 STAL	1.000 -> 1.000 S (0.041)	1.000 -> 1.000 S (0.019)
601	1.000 S	0.949 STAL	1.000 -> 1.000 S (0.030)	1.000 -> 1.000 S (0.042)
602	1.000 S	0.853 STAL	1.000 -> 1.000 S (0.154)	1.000 -> 1.000 S (0.072)
603	1.000 S	0.951 STAL	1.000 -> 1.000 S (0.046)	1.000 -> 1.000 S (0.017)
604	1.000 S	0.907 STAL	1.000 -> 1.000 S (0.024)	1.000 -> 1.000 S (0.039)
605	1.000 S	0.930 STAL	1.000 -> 1.000 S (0.032)	1.000 -> 1.000 S (0.028)
606	1.000 S	0.950 STAL	1.000 -> 1.000 S (0.031)	1.000 -> 1.000 S (0.112)
607	1.000 S	0.922 STAL	1.000 -> 1.000 S (0.036)	1.000 -> 1.000 S (0.015)

Registered: from scratch 0.921 vs 1.000, -0.079 (permutation p 0.0002), 0/8 vs 8/8 (Fisher p 0.0002); constructed |.| HELD (8/8 SOLVED, 0/8 LOST).

Observation-step-free torus (`GENERALITY_ANALYSIS.txt`):

seed	signed	|.|	constructed signed (lr 1e-4)	constructed |.| (lr 1e-4): peak loss, epoch of peak / of re-entry
700	1.000 S	0.874 STAL	1.000 S	1.000 S; 0.168, 29 / 153
701	1.000 S	0.707 STAL	1.000 S	1.000 S; 0.110, 40 / 121
702	1.000 S	0.946 DESC	1.000 S	1.000 S; 0.173, 19 / 204
703	1.000 S	0.879 STAL	1.000 S	1.000 S; 0.143, 52 / 149
704	1.000 S	0.892 STAL	1.000 S	1.000 S; 0.131, 13 / 119
705	1.000 S	0.907 STAL	1.000 S	1.000 S; 0.155, 13 / 148
706	1.000 S	0.877 STAL	1.000 S	1.000 S; 0.208, 36 / 164
707	1.000 S	0.960 DESC	1.000 S	1.000 S; 0.117, 23 / 139

Registered: from scratch 0.880 vs 1.000, -0.120 (p 0.0002), 0/8 vs 8/8; the constructed |.| model leaves the solved region on 6/8 seeds (peak loss 0.11-0.21; signed control 0.004-0.017) and ends SOLVED on 8/8 at 1.000: by the reading fixed before launch, perturbed and re-found.

Recipe and scale (`ROBUST_ANALYSIS.txt`, seeds 1400-1407; accuracy at the training length, class):

```
ring  seed | def:SO | def:AO | wd0:SO | wd0:AO | lrx3:SO | lrx3:AO | lrd3:SO | lrd3:AO
1400 | 1.0000 SOLV | 0.8230 STAL | 1.0000 SOLV | 0.9078 DESC | 1.0000 SOLV | 1.0000 DESC |
1.0000 SOLV | 0.8893 STAL
1401 | 1.0000 SOLV | 0.9155 STAL | 1.0000 SOLV | 0.6239 STAL | 1.0000 SOLV | 0.9596 STAL |
0.6194 STAL | 0.6221 STAL
1402 | 1.0000 SOLV | 0.9237 STAL | 1.0000 SOLV | 0.8878 STAL | 1.0000 SOLV | 0.9702 STAL |
1.0000 SOLV | 0.8319 STAL
1403 | 1.0000 SOLV | 0.9136 STAL | 1.0000 SOLV | 0.8800 STAL | 1.0000 SOLV | 0.9995 SOLV |
1.0000 SOLV | 0.9167 STAL
1404 | 1.0000 SOLV | 0.9118 STAL | 1.0000 SOLV | 0.9119 STAL | 1.0000 SOLV | 1.0000 SOLV |
1.0000 SOLV | 0.8313 STAL
1405 | 1.0000 SOLV | 0.8791 STAL | 1.0000 SOLV | 0.8858 STAL | 1.0000 SOLV | 0.9650 STAL |
1.0000 SOLV | 0.8526 STAL
1406 | 1.0000 SOLV | 0.9211 DESC | 1.0000 SOLV | 0.9110 STAL | 1.0000 SOLV | 1.0000 SOLV |
0.9794 STAL | 0.9062 STAL
1407 | 1.0000 SOLV | 0.9997 SOLV | 1.0000 SOLV | 0.9735 DESC | 1.0000 SOLV | 0.9998 SOLV |
1.0000 SOLV | 0.8801 STAL
torus 1400 SO 1.0000 SOLV | AO 0.9941 DESC     1404 SO 1.0000 SOLV | AO 0.9935 SOLV
      1401 SO 1.0000 SOLV | AO 0.9981 SOLV     1405 SO 1.0000 SOLV | AO 0.9706 DESC
      1402 SO 0.9941 SOLV | AO 0.9705 DESC     1406 SO 1.0000 SOLV | AO 0.9098 DESC
      1403 SO 1.0000 SOLV | AO 0.9454 DESC     1407 SO 1.0000 SOLV | AO 0.9544 DESC
```

Learning-rate sweep (`LRSWEEP_RESULTS.md`, seeds 1500-1507): per-seed table in H2.

C6. Text world (Sec. 7; `TEXTWORLD_ANALYSIS.txt`)

seed	path, 1 layer	index, 1 layer	index, 2 layers
0 (pilot)	0.881	0.504	0.695
1 (pilot)	0.998	0.504	0.723
2	0.999	0.503	0.927
3	0.995	0.506	0.722
4	0.999	0.505	0.926
5	1.000	0.506	0.706
6	1.000	0.506	0.760
7	0.882	0.504	0.720

Fresh seeds 2-7 (`docs/audits/2026-09-27/fresh_seeds.txt`): path 0.979 +/- 0.048 (6/6 SOLVED), index 1 layer 0.505 (0/6), index 2 layers 0.793 (0/6); path - index one layer +0.474, permutation and Fisher p 0.0022. All eight seeds: path 0.969 +/- 0.054 (7/8; seed 0 still descending), +0.464 (p 0.0002; Fisher p 0.0014); path - index two layers +0.197 (p 0.0009).

C7. Dyck-2

Per-seed values are in `DYCK_MDEPTH_RESULTS.json` (cells `T12|{RoPE,PoPE,MapWM,MapPoPE}|{1-4}L|L32D12`); every one of the eight seeds has a positive position effect at every depth (sign-flip p 0.0078 per depth).

C8. SOLVED-threshold sensitivity

Sign batches (`docs/audits/2026-10-15/reanalyses/solved_thresholds_out.md`; predictions and decision rules written before the re-analysis; two-sided Fisher p in brackets):

batch	contrast	loss < 0.02	loss < 0.05 (registered)	loss < 0.1	accuracy >= 0.99
SIGN_MATCHED torus	|.| vs signed	0/8 vs 8/8 (0.0002)	0/8 vs 8/8 (0.0002)	0/8 vs 8/8 (0.0002)	0/8 vs 7/8 (0.0014)
SIGN_MATCHED torus	softplus vs signed	0/8 vs 8/8 (0.0002)	1/8 vs 8/8 (0.0014)	1/8 vs 8/8 (0.0014)	1/8 vs 7/8 (0.0101)
RING_DIAL p 0.5	|.| vs signed	2/8 vs 8/8 (0.0070)	2/8 vs 8/8 (0.0070)	4/8 vs 8/8 (0.0769)	2/8 vs 8/8 (0.0070)
RING_DIAL p 0.75	|.| vs signed	1/8 vs 8/8 (0.0014)	1/8 vs 8/8 (0.0014)	2/8 vs 8/8 (0.0070)	2/8 vs 8/8 (0.0070)
WRAP_EXISTENCE ring	|.| vs signed	0/8 vs 8/8 (0.0002)	0/8 vs 8/8 (0.0002)	0/8 vs 8/8 (0.0002)	0/8 vs 8/8 (0.0002)
MECHANISM ring	|.| vs signed, same-direction start	0/8 vs 7/8 (0.0014)	0/8 vs 7/8 (0.0014)	0/8 vs 7/8 (0.0014)	0/8 vs 7/8 (0.0014)
ACCOUNTS T2 ring	frozen vs free move latents	1/8 vs 7/8 (0.0101)	1/8 vs 7/8 (0.0101)	1/8 vs 7/8 (0.0101)	1/8 vs 7/8 (0.0101)
GENERALITY torus	|.| vs signed	0/8 vs 8/8 (0.0002)	0/8 vs 8/8 (0.0002)	0/8 vs 8/8 (0.0002)	0/8 vs 8/8 (0.0002)
GENERALITY ring	CoPE vs signed	0/8 vs 8/8 (0.0002)	0/8 vs 8/8 (0.0002)	1/8 vs 8/8 (0.0014)	0/8 vs 8/8 (0.0002)

ROBUST (declared secondary, loss < 0.1): ring default 1/8 vs 8/8 (p 0.0014), weight decay 0 1/8 vs 8/8 (0.0014), lr / 3 0/8 vs 7/8 (0.0014), lr x3 4/8 vs 8/8 (0.077), torus 2 layers d 256 5/8 vs 8/8 (0.20); on that torus cell the accuracy contrast (0.967 vs 0.999, p 0.0005) carries the result. LRSWEEP (loss < 0.1, selection redone): 2/8 vs 8/8, same label.

Rank batches (analysis; read from the per-seed final-5% losses in C4): at loss < 0.02 the counts are 0 / 2 / 2 / 6 / 8 / 8 of 8 (shared r2, per-head r2, block-diagonal r4, per-head r3, shared r4, per-head r4), identical to the registered 0.05; at loss < 0.1 they are 1 / 2 / 2 / 6 / 8 / 8 (shared r2 seed 1 at 0.051 enters).

C9. Attention-dropout evaluation scale

Models trained with attention dropout expect the 1/(1 - p) attention scale; evaluated with it, below-ceiling one-layer runs score higher. The registered text-world evaluation re-run with the scale (`docs/audits/2026-10-04/DROPOUT_RESCORE.md`, all 24 runs) gives path 0.969 -> 0.993, index one layer 0.505 -> 0.506, index two layers 0.772 -> 0.772; path - index one layer +0.464 -> +0.488 (p 0.0002). The re-score was a declared secondary of RING_DIAL, RING_BUDGET, WRAP_EXISTENCE, GENERALITY, MECHANISM and WRAP_COPE and changed no label there; it was applied to SIGN_MATCHED by the same audit with no verdict changed. ACCOUNTS, TORUS_START, GAUGE and ROBUST were not re-scored (the correction does not apply at two layers). All main-text accuracies are eval mode without the scale.

D. Constructions

D1. The rank-2 projection (Sec. 6.1; `RANK_PROJ_RESULTS.md`)

Each solved shared rank-4 model (seeds 0-7, a stored batch) is projected onto the top two directions of its action latent over the vocabulary (top-2 spectral energy 0.99952-0.99994), every other weight kept, and installed in the shared rank-2 architecture. Frozen, the projections score 0.9955 at T = 1024 against their sources' 0.9966 (at least 0.981 on every seed; per seed 0.9993, 1.0000, 0.9988, 1.0000, 0.9850, 0.9806, 1.0000, 1.0000). Trained 900 epochs from a learning-rate restart at 1e-3, 7/8 are SOLVED (one DESCENDING), as for the rank-4 control trained from its own solution (7/8); rank 2 continued from its own 900-epoch weights solves 1/8. Registered verdict: STABLE.

D2. Per channel (Lemma P1.0 in v3)

For a signed step Delta and omega != 0, Delta' = Delta mod (2 pi / |omega|) lies in [0, 2 pi / |omega|) and omega Delta' = omega Delta (mod 2 pi) for either sign of omega. The model reads each channel's angle only through cos and sin, so the model with Delta' is the same function.

D3. The balanced |.| construction (Proposition 1(a); the registered existence test)

Given a solved signed observation-step-free model with m stepping move tokens (m = 2 ring, 4 torus; r = 4):

1. Re-solve W_in so that the move latents are c e_1, ..., c e_m (c = the source's mean move-latent norm). W_in acts only on the move embeddings, so this is a free gauge.
2. Per channel set omega' = max(|omega|, 1) and the steps to remainder(omega Delta, 2 pi) / omega' (angles in [0, 2 pi) for either sign of omega).
3. Set W_out[:, j] = step_j / c for the m move columns and 0 elsewhere; copy every other weight.

The signed control is built by the same function with the unwrapped signed angle (steps omega Delta / omega'), so the two differ only in the wrap. On the torus m = 4 = r, exactly at Proposition 1(a)'s limit.

Exactness. Ring: frozen accuracy = source on 8/8; maximum angle error 1.2e-6 rad (|.|), 6.1e-7 (signed); maximum logit difference 5.1e-4 at T = 128. Torus: frozen = source on 8/8; maximum angle error 1.4e-6 rad (|.|), 5.8e-7 (signed). *Weight scale.* Ring: maximum |W_out| 0.31-0.38 in the signed sources, 1.07-1.70 built; |W_in| 0.15-0.21 -> 0.19-0.33. Torus: maximum |W_out| 0.48-0.58 -> 1.23-1.55, maximum |omega| 0.68-0.96 -> 1.00. Within about 5x of the trained scale.

Why the scale matters. A construction that keeps the source's W_in and solves W_out at minimum norm is also exact, but in a solved signed model the two move latents are nearly antiparallel (cosine -0.987 to -0.998), so W_out must be huge (up to 16,320 against 0.31-0.38). Adam's per-entry steps then lose that solution in the warm-up (8/8 LOST)

before re-finding it (7/8 SOLVED). An existence test is informative only near a trained scale, which the balanced construction provides.

D4. The full |.| arms

To show that the task, not only a given signed table, is representable by a full |.| model (observations stepping), W_in is chosen to project onto two directions orthogonal to all 17 observation embeddings and W_out is solved per channel (`docs/audits/2026-10-12/ring_dial_audit.md` M1). In the real modules: |.| steps >= 0, observation-token |Delta| <= 1.7e-6, cos / sin equal to the signed code's to 6.3e-4 over a 512-step walk, circular mismatch <= 4.6e-4 on 2,000 same-cell revisit pairs.

D5. Softplus (Proposition 1(f))

Softplus cannot reach 0, but observation steps need only be 0 mod 2 pi. With W_in annihilating every observation embedding, each observation steps softplus(0) = ln 2 in every channel, and omega'_c = 2 pi / ln 2 makes that exactly one turn; the move columns are set to softplus^{-1}((remainder(Theta, 2 pi) + 2 pi) / omega'). Built in the real softplus model from a solved ring model (`docs/paper/review_draft5/theory_check_extra.py` (2), CPU): minimum step 0.693 > 0, angle error mod 2 pi 8.4e-15 rad over all 19 tokens, logits equal to the source's to 2.8e-11 (logit scale 21.6). On the torus m = 4 = r, so the same construction applies.

D6. Continuations and the LOST rule

A constructed model is trained further on a fresh walk stream and compared with its signed control at the same seed. *LOST* (registered): the run's maximum per-epoch training loss during the continuation exceeds 0.3, or exceeds 10x its control's maximum at that seed and 0.05. *HELD* = SOLVED on at least n - 1 seeds, mean accuracy at least control - 0.01, and LOST on at most one seed. Ring (lr 1e-3): HELD, 8/8 SOLVED, 0/8 LOST (maximum per-epoch loss 0.024-0.154 against the control's 0.015-0.112). Torus (lr 1e-4): LOST on 6/8 and SOLVED on 8/8 at the end, read by the rule fixed before launch as perturbed and re-found; at lr 1e-3 (descriptive) 6/8 SOLVED (0.988).

D7. What the held model keeps (ACCOUNTS T1, registered)

Weighted by usage (mean query norm at move tokens times mean key norm at observation tokens, per channel), the held ring |.| model keeps its cancelling code from 0.99 at construction to 0.98 after training; unweighted (every channel equal) the share falls from 0.87 to 0.18, because channels attention does not read leave the code. Weight decay 0 vs 0.05 changes nothing detected (-0.003).

E. Theory

Labels: main-text Proposition 1 = P1(a) and P1(f) of `APPENDIX_theory_v3.md`; Lemma 1 = T1; Proposition 2 = P6; Propositions 3 and 4 below = P3 and P4. Every check script is in `docs/paper/appendix_checks/` (CPU, seconds each).

Notation. A rotary channel is c = (h, b), head h and block b, acting on coordinates (2b, 2b + 1) of head h; R(phi) is the 2D rotation. The phase is theta_{t,c} = omega_c sum_{u <= t} S_{c, x_u}, with S = W_out W_in E the C x V step table (rank at most r) and Theta = diag(omega) S the angle table. omega is free in sign. For an inverse pair of tokens (x, y), channel c *cancels through zero* if Theta_{c,x} + Theta_{c,y} = 0 with Theta_{c,x} != 0, *cancels by wrapping* if the sum is a non-zero multiple of 2 pi, and is *blind* if both angles are 0 mod 2 pi.

E1. Score identity (A0)

With alpha piecewise constant, alpha = Delta_u on (u - 1, u], the homogeneous part of Howard et al.'s integrator, dF/dt = -s alpha(t) F, has propagator F(t) = exp(-s (Phi_t - Phi_{t'})) F(t'). With s = i omega, channel c of the rotary score is q_c^T R(theta_s - theta_t) k_c = Re(conj(q~*c) k~_c exp(-i omega_c (Phi_t - Phi_s))), with q~ = q*{2b} + i q_{2b+1}. *Proof.* R(theta) acts on q~ as multiplication by e^{i theta}; flows over tokens compose by adding exponents. The identity for complex SSMs is Mamba-3's Proposition 3; ours is only its mapping onto Howard's three rate cases. Checked (`a0_propagator.py`): 3e-14 (propagator), 4e-15 (score).

E2. Proposition 1 (representability by wrapping; P1(a), P1(f))

(a) Let S be a signed table with at most m <= r non-zero columns (stepping tokens x_1, ..., x_m), whose embeddings are linearly independent modulo the span of the other embeddings (generic when V <= d_model, here 19 or 21 <= 128; automatic in observation-step-free models). Then a |.| model with the same embeddings and content weights has an angle table equal to Theta mod 2 pi, entrywise.

Proof. Pick W_in' with W_in' e_{x_j} = u_j (the j-th unit vector of R^r) and W_in' e_y = 0 for every other token. Set omega' = max(|omega|, 1) per channel and W_out'[:, j] = remainder(Theta_{., x_j}, 2 pi) / omega' >= 0, other columns zero. Then W_out' W_in' E is entrywise non-negative, so |.| leaves it unchanged, and it reproduces each stepping column mod 2 pi. [] *Full arms:* choosing W_in' to annihilate every observation embedding leaves only the moves stepping, so every |.| model in the paper can represent any code in which only the moves step, including the map codes of the trained signed observation-step-free models. Checked (`p1_wrap.py`): angle error <= 7e-14, logits <= 8e-14.

(f) Softplus. Under the same assumptions, send the stepping embeddings to u_1, ..., u_m and every other embedding to 0, set omega'c = 2 pi / ln 2 for every c, and W_out'[c, j] = softplus^{-1}((remainder(Theta{c,x_j}, 2 pi) + 2 pi) / omega'_c). Every non-stepping token then steps softplus(0) = ln 2, an angle of exactly 2 pi, and every stepping token's angle is Theta mod 2 pi.

Proof. softplus is a bijection from R onto (0, inf); the targets lie in [ln 2, 2 ln 2); omega' ln 2 = 2 pi. [] Checked (`theory_check_extra.py` (2)): D5.

E3. Lemma 1 (relative-sign gauge; T1)

Take a MapWM model with any number of layers and a set F of channels. Form a second model by (a) negating every token's step in each channel of F (by negating omega_c; in the signed class equivalently row c of W_out, which is not available in the |.| class since |.|

is applied after W_out) and (b) negating output coordinate h d_head + 2b + 1 of the query and key projections (weight row and bias entry, every layer) for each c = (h, b) in F. The two models have identical attention scores at every layer, hence identical logits.

Proof. With J = diag(1, -1), J R(phi) J = R(-phi) and J is orthogonal. (a) replaces theta_{t,c} by -theta_{t,c}; (b) replaces q_c, k_c by J q_c, J k_c, exactly including biases, since nothing between the projection and the rotation mixes coordinates. The channel term becomes (J q_c)^T R(-(theta_s - theta_t)) (J k_c) = q_c^T R(theta_s - theta_t) k_c. All other components are untouched, and every layer uses the same cos / sin, so the scores agree layer by layer. [] Checked (`t1_gauge.py`): 35 channels flipped and reflected give bitwise-identical logits (control without the reflection: 6.6).

Corollaries. (i) The |.| class realises a signed angle table Theta exactly as real numbers, with the same latent, iff every row of Theta is sign-coherent (all stepping tokens in a channel step the same way); omega carries each channel's direction. So the |.| constraint removes only the relative signs of the tokens that step in a channel. (ii) It therefore excludes no map *function* for which a code with sign-coherent rows exists, and that code needs no wrapping: one stepping token (parity, Z_m counting), or two moves that step the same way (+1 and +2: a(+2) = 2 a(+1)). Other codes of the same function can still be excluded as parameter points: on +1/+2 the code a(+1) = 1, a(+2) = 2 - 2 pi has an opposite-sign row. (iii) Up to the gauge, the excluded tables are exactly those with a row containing entries of strictly opposite sign. For an inverse pair, every through-zero cancelling channel is such a row; cancellation by wrapping, (a, 2 pi - a), is allowed and is representable whenever Proposition 1 applies. The constraint removes the through-zero branch of the inverse-pair problem, not cancellation.

E4. One channel's geometry (P3, P4)

Take one channel and an inverse pair (x, y) with angles (a_x, a_y), everything else fixed; the loss is a function on the angle torus. Write sigma = a_x + a_y.

Proposition 3 (cut circle). (a) The closed quadrant [0, inf)^2 maps onto the torus, so the signed and |.| classes attain the same loss values in this channel. (b) The non-blind cancelling set in the quadrant consists of the segments sigma = 2 pi k, k >= 1; the through-zero line meets the quadrant only at the blind origin. (c) From a start in the open quadrant with sigma < 2 pi, the infimum distance (in angle coordinates) to a non-blind cancelling point is min(sigma, 2 pi - sigma) / sqrt 2 in the signed class and (2 pi - sigma) / sqrt 2 in the |.| class; they coincide iff sigma >= pi. (d) Any continuous |.| path from |sigma| < pi to a non-blind cancelling point passes through |sigma| = pi.

Proof. (a), (b) from the quotient map. (c) The distance to sigma = 2 pi k is |sigma - 2 pi k| / sqrt 2; the |.| class has only k >= 1 (with omega < 0 also the negative quadrant, which is farther). (d) |sigma| is continuous along a |.| path and a non-blind cancelling point has |sigma| >= 2 pi; intermediate value theorem. [] Checked (`p3_p5_geometry.py`): a brute-force grid matches both formulas to 2e-4.

Proposition 4 (sticky wall). Let s be the pre-|.| step of one token in one channel and a = omega |s|, with L(s) = l(|s|) and l continuously differentiable near 0. (a) For s != 0, dL/ds = omega sign(s) dL/da. (b) If l'(0) > 0, s = 0 is a strict local minimum of L along s with every other parameter fixed: L(s) - L(0) = l'(0) |s| + o(|s|). (c) Gradient descent on s with

step eta, l' in [mu, M] on [0, rho], mu > 0, eta M <= rho and |s_0| <= rho, gives |s_{k+1}| <= max(|s_k| - eta mu, eta M): the iterate chatters within eta M of the wall.

Proof. (a) chain rule for |.|; (b) first-order expansion; (c) case split on |s_k| >= eta l'. [] Proposition 4 is a statement along one coordinate; in parameter space s = w_c . z_x is shared across tokens and channels.

E5. Proposition 2 (CARoPE cannot cancel; P6)

CARoPE (faithful, `model_mono_baselines.MapFormerWM_CARoPEF`) advances channel i of head h by f_h(x)^i per token, with f in (0, 1), so every per-token advance lies in (0, 1]; channel 0 advances by exactly 1, an index clock.

(a) The phase of any interval of k >= 1 tokens lies in (0, k]. The shortest return (out and back) spans 3 tokens between the move query and the observation key, and 4 between two observation tokens, so its phase lies in (0, 4] subset (0, 2 pi): no channel, at any parameter value, reads every return at phase 0. (b) Fix eps in (0, pi - 2). If a channel reads each return of an excursion of J moves out and J back within eps of 0 mod 2 pi, then the move-driven phase of any interval with at most 2J moves lies in [0, 2 eps]: every channel is either a clock (it misses some short return) or blind over that range.

Proof. (a) Sum k numbers in (0, 1]. (b) The depth-j return phase P_j satisfies P_1 in (0, 3] and P_{j+1} - P_j in (0, 4]; the arc (eps, 2 pi - eps) is longer than 4 when eps < pi - 2, so a sequence starting in (0, eps] with increments at most 4 cannot jump it; induction gives P_j in (0, eps], hence J (c_+ + c_-) <= eps. [] Checked (`p6_carope.py`): eight trained ring CARoPE models; 10,331 random draws satisfy the bound.

F. Exchange-rate details

Dyck-2 (`DYCK_MDEPTH_RESULTS.md`). Task and metric as MapFormer v4 App. B.4, trained and scored at length L = 32 and nesting depth D = 12; primary metric A2f, Hewitt's distance-averaged closing accuracy over prefixes where a closer can be emitted (CE-optimal predictor 1.000, chance 0.500, best stack-free floor 0.594). 2 heads, d_model 128, 1-4 layers; path-integrated arms MapWM and MapPoPE at rank 2; index arms RoPE and PoPE at base 10000 (and base 32 in the 4-layer 3x cell); n = 8 paired seeds. Position main effect = ((MapWM - RoPE) + (MapPoPE - PoPE)) / 2.

condition	layers	RoPE	PoPE	MapWM	MapPoPE	position main effect
1x budget	1	0.648	0.640	0.996	0.998	+0.353
1x budget	2	0.889	0.850	1.000	1.000	+0.130
1x budget	3	0.964	0.946	1.000	1.000	+0.045
1x budget	4	0.982	0.971	1.000	1.000	+0.024
3x budget	4	0.998	0.997	1.000	1.000	+0.002
mixture of depths 4-12	4	0.898	0.881	0.999	0.999	+0.110

Every effect is positive on 8/8 seeds (sign-flip p 0.0078). Index base 32 minus base 10000 at four layers, 3x budget: +0.001 (RoPE), -0.001 (PoPE), within the MDE. Three times the budget raises the index arms by +0.021 at four layers (8/8).

Ring (C3). The index-one-layer accuracies are not monotone in p_plus (0.710, 0.677, 0.824, 1.000 on fresh seeds; every pair differs, p 0.0022-0.0108), so the exchange rate, not the one-layer accuracy, is the quantity that tracks whether steps are undone.

Torus (`FACTORIAL_RESULTS.md`, Fig. F1). Strata (short gap / long gap / wrap-only; mean and run range, n = 4): path one layer 1.000 / 1.000 / 1.000; index one layer 0.769 (0.735-0.787) / 0.480 (0.456-0.508) / 0.482 (0.470-0.495); four layers 0.943 (0.900-0.958) / 0.458 (0.367-0.506) / 0.457 (0.381-0.494); looped x4 0.938 (0.937-0.940) / 0.507 (0.496-0.531) / 0.509 (0.496-0.521). Floors on this stream (analysis): constant 0.508 / 0.515 / 0.517, retrace 0.912 on short gaps. Memorisation check: four index layers 0.897 on their own training map against 0.860 held out; looped 0.862 against 0.864; path 1.000 on both.

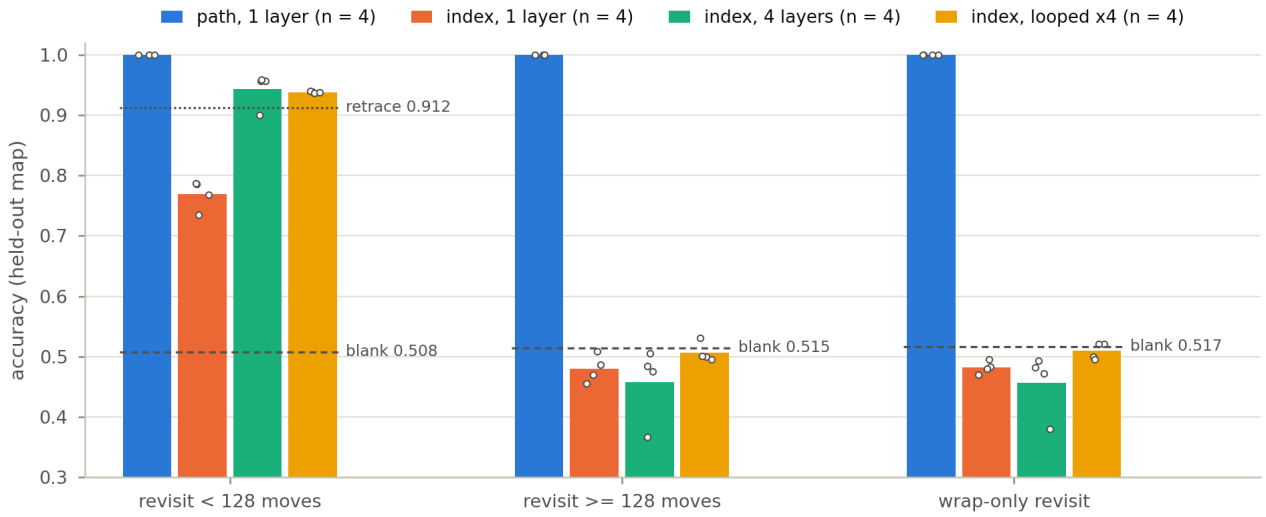


Figure F1 (`fig_factorial_strata`). Revisit accuracy by stratum on the torus at T = 1024 (trained and tested at T = 1024; bars: means of n = 4 runs, dots: runs). Dashed: always-blank within each stratum, 0.508 / 0.515 / 0.517; dotted: retrace within the short-gap stratum, 0.912 (elsewhere it equals the constant); floors on this batch's own evaluation walks. Path integration with one layer is at 1.000 in every stratum. Four index layers or a loop of four lift short-gap revisits to 0.94, just above retrace; at long gaps and wrap-only revisits every index model stays at or below the constant (0.46-0.51).

G. Rank details

Arms (A1). Shared r2 (`Vanilla`), per-head r2 (`Vanilla_r2ph`), block-diagonal r4 (`Vanilla_r4mibd`: the shared rank-4 latent with W_out's off-diagonal blocks held at zero), per-head r3 (`Vanilla_r3ph`), shared r4 (`Vanilla_r4mi`), per-head r4 (`Vanilla_r4ph`), all built from the shared rank-2 model's initial weights at each seed. Initial angle-increment scale (std): 0.208 for block-diagonal r4 against 0.33-0.34 for the other arms, which fail (rank 2) or solve (rank 4) from the same scale. Per-seed results: C4.

Training curves (Fig. G1; `fig_loss_curves_rank_out.txt`). Shared r2 0/8 SOLVED (4 STALLED, 4 DESCENDING; median final loss 0.35) against shared r4 8/8 (2.8e-3); per-head r2 2/8 (median 0.22), r3 6/8 (3.5e-3), r4 8/8 (3.1e-3). The rank-3 and rank-4 medians cross the threshold at epochs 478-532; the rank-2 medians never do.

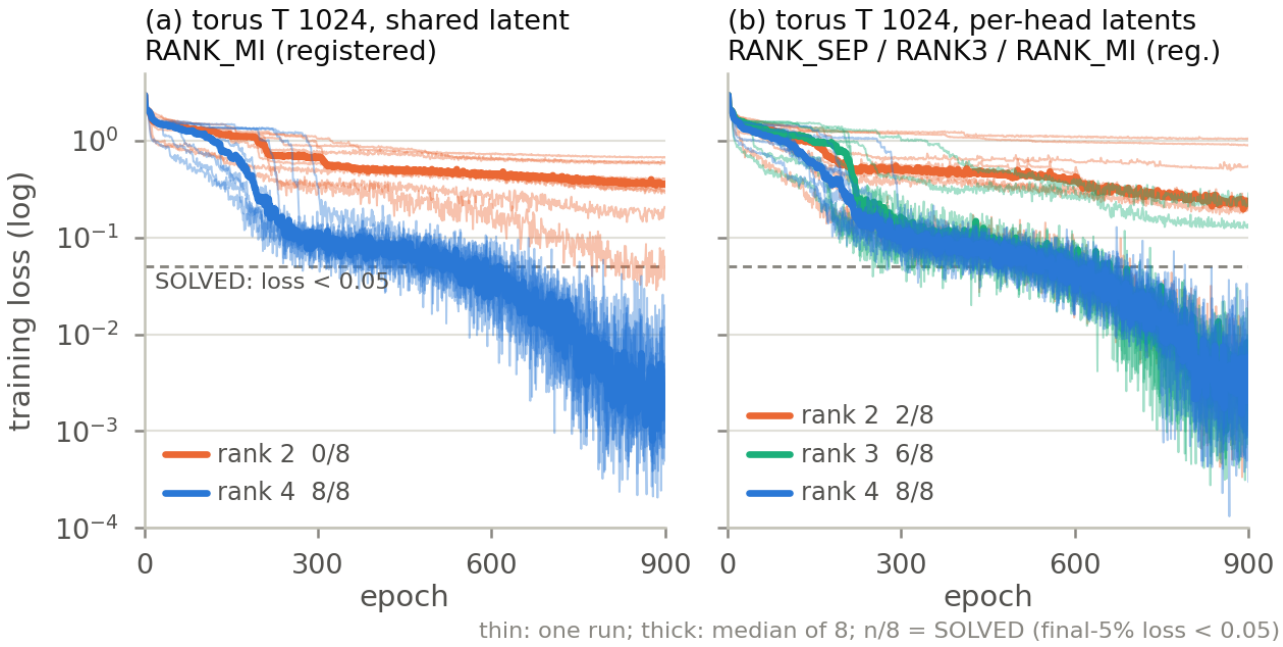


Figure G1 (`fig_loss_curves_rank`). Training loss per epoch on the torus (T = 1024) by the per-head rank of the content-to-angle map; thin lines are runs, thick lines medians of 8; dashed: the SOLVED threshold. (a) One latent shared by the heads: rank 2 solves 0/8, rank 4 8/8. (b) Per-head latents: rank 2 2/8, rank 3 6/8, rank 4 8/8.

Restart curves (`RANK_PROJ_ANALYSIS.txt`). Projected rank 2: peak training loss 0.37-0.46 after the restart, back below 0.05 at epochs 553-654 on 7/8 seeds. Rank-4 control: peak 0.26-0.30, back at 497-669 on 7/8. Rank 2 continued: peak 0.55-0.82, back on 1/8.

Rank against dimension (`RANK_ND_RESULTS.md`, `RANK_WRAP_RESULTS.md`, `RANK_TESTS_RESULTS.md`). Per-head rank, every arm from the same base, trained and tested at T = 1024, 900 epochs:

torus	D	per-head rank	wrap-only share of revisits	SOLVED	held-out accuracy
32 per side	2	2	0.37	1/8	0.769
32 per side	2	3	0.37	8/8	0.999
10 per side	3	3	0.71	1/8	0.800
10 per side	3	4	0.71	4/8	0.847
18 per side	3	4	0.35	8/8	1.000

2D rank 3 vs rank 2: 8/8 vs 1/8 (Fisher p 0.0014), +0.230 (permutation p 0.0014). 3D, 18 vs 10 per side at rank 4: +0.153 (p 0.0085). On a 100-cell 2D torus (10 per side) rank 3 memorises its training map (0.986 on it, 0.273 on an unseen map), so that cell does not measure map learning. 1D ring (32 cells, T = 128, 300 epochs): rank 1 7/8 (0.993), rank 2 8/8 (1.000); with the component common to the move steps removed by construction, rank 1 solves 8/8 in a median of 24 epochs (58 without).

Escape test (`RANK_TESTS_RESULTS.md`). Lifting the eight stalled shared-rank-2 runs to rank 4 (function-preserving) and training 600 epochs solves 3/8, against 1/8 for rank 2 continued

and 2/8 for a fresh rank-4 step map on the same networks; at this budget the contrasts are not resolved.

H. Sign details

H1. Learning rates (`LRSWEEP_RESULTS.md`; ring, observation-step-free, seeds 1500-1507)

learning rate	signed SOLVED, accuracy	|.| SOLVED, accuracy
1e-3	8/8, 1.000	1/8, 0.917
3e-3	8/8, 1.000	2/8, 0.963
1e-2	4/8, 0.972	1/8, 0.872
3e-2	1/8, 0.559	0/8, 0.584

Registered B1 (each arm at its best rate, selection redone in the null over all 12,870 relabellings): |.| 3e-3 2/8 vs signed 1e-3 8/8, -0.037, decision p 0.0070 (SOLVED) / 0.0057 (accuracy). The |.| optimum is interior to the grid. Three |.| runs at 3e-3 were still descending at 300 epochs. At 3e-2 both arms sit near the 0.50 blank floor. Per seed:

```
seed | 1e-3:SO | 1e-3:AO | 3e-3:SO | 3e-3:AO | 1e-2:SO | 1e-2:AO | 3e-2:SO | 3e-2:AO
1500 | 1.0000 SOLV | 0.9362 STAL | 1.0000 SOLV | 0.9111 STAL | 0.9460 DESC | 0.9936 DESC |
0.4960 STAL | 0.4958 STAL
1501 | 1.0000 SOLV | 0.8759 STAL | 1.0000 SOLV | 0.9662 DESC | 0.9460 STAL | 0.4960 STAL |
0.4960 STAL | 0.4960 STAL
1502 | 1.0000 SOLV | 0.8995 STAL | 1.0000 SOLV | 0.9428 STAL | 0.9407 STAL | 0.9703 DESC |
0.4960 STAL | 0.4960 STAL
1503 | 1.0000 SOLV | 0.9116 STAL | 1.0000 SOLV | 1.0000 SOLV | 1.0000 SOLV | 0.6000 STAL |
0.4960 STAL | 0.4960 STAL
1504 | 1.0000 SOLV | 0.9415 STAL | 1.0000 SOLV | 0.9686 STAL | 0.9460 STAL | 1.0000 DESC |
0.4960 STAL | 0.8159 DESC
1505 | 1.0000 SOLV | 1.0000 SOLV | 1.0000 SOLV | 0.9993 SOLV | 1.0000 SOLV | 0.9950 DESC |
0.9998 SOLV | 0.4961 STAL
1506 | 1.0000 SOLV | 0.8736 STAL | 1.0000 SOLV | 0.9595 DESC | 1.0000 SOLV | 0.9996 SOLV |
0.4960 STAL | 0.4960 STAL
1507 | 1.0000 SOLV | 0.8971 STAL | 1.0000 SOLV | 0.9587 DESC | 1.0000 SOLV | 0.9184 DESC |
0.4959 DESC | 0.8799 DESC
```

H2. Recipe cells (`ROBUST_RESULTS.md`, seeds 1400-1407)

cell	recipe	|.| accuracy, SOLVED	signed accuracy, SOLVED	d [95% CI]	perm p	Fisher p
ring	lr 1e-3, weight decay 0.05	0.911, 1/8	1.000, 8/8	-0.089 [-0.125, -0.055]	0.0002	0.0014
ring	lr 1e-3, weight decay 0	0.873, 0/8	1.000, 8/8	-0.127 [-0.200, -0.075]	0.0002	0.0002
ring	lr 3e-3, weight decay 0.05	0.987, 4/8	1.000, 8/8	-0.013 [-0.025, -0.005]	0.026	0.077
ring	lr 3.3e-4, weight decay 0.05	0.841, 0/8	0.950, 6/8	-0.109 [-0.215, +0.010]	0.089	0.0070
torus		0.967, 2/8	0.999, 8/8		0.0005	0.0070

cell	recipe	|.| accuracy, SOLVED	signed accuracy, SOLVED	d [95% CI]	perm p	Fisher p
	2 layers, d_model 256			-0.032 [-0.055, -0.015]		

At lr 3e-3 the registered contrast was below its decision threshold in this batch (4/8 vs 8/8); on fresh seeds in the sweep the same rate gives 2/8 vs 8/8 (H1). Declared secondaries: cancelling share (unweighted) |.| 0.04-0.13 in every cell, signed 0.43-1.00; torus strata (short / long / wrap-only) signed 1.000 / 0.999 / 0.992, |.| 0.989 / 0.911 / 0.799; on the torus 6/8 |.| runs were still descending at 900 epochs (final losses 0.018-0.144).

H3. Starts (`MECHANISM_RESULTS.md`, ring p_plus 0.5, seeds 800-807; `TORUS_START_RESULTS.md`, torus, seeds 1200-1207)

Observation-step-free models differing only in the initial move angles (every other weight identical at a seed):

initial move angles	signed (ring)	|.| (ring)	signed (torus)	|.| (torus)
default (small, signs independent)	8/8	0/8	8/8	1/8
the |.| start: both moves rotating the same way	7/8	0/8	--	--
large, opposite-move sums below pi, U(0.9, 1.5)	--	2/8	--	0/8
large, sums above pi, U(1.7, 4.5)	8/8	8/8	2/8	0/8

Ring, registered: same start, sign matters (0/8 vs 7/8, Fisher p 0.0014; -0.092 [-0.135, -0.045]); a start past pi rescues |.| (8/8 vs 0/8, +0.109 [+0.085, +0.135], p 0.0002); a large start with sums below pi does not (2/8; vs past pi Fisher p 0.0070); from past pi the sign makes no difference (8/8 vs 8/8). The large start also moves the angles 13-22x faster per Adam step (the sub-pi control about 5x), so sum and size are only partly separated. Analysis: signed models from the small starts end with 98-99% of channels opposite-signed and cancel through zero; from past pi both signs cancel by wrapping (signed 88%, |.| 100% of cancelling channels); the rescued |.| models cancel in 23% of channels. Torus, registered: the past-pi start does not rescue |.| (0/8 vs 1/8, +0.005 [-0.060, +0.070]), and it slows the signed model (2/8; 5/8 runs per large-start arm still descending at 900 epochs); the default-start sign cost replicates (1/8 vs 8/8, Fisher p 0.0014). The start lever is a ring result.

H4. Moves +1/-1 vs +1/+2 (`GAUGE_RESULTS.md`, ring, seeds 1300-1307)

By Lemma 1 (ii) a |.| step represents, without wrapping or crossing zero, a map whose two moves step the same way. With moves +1/-1 (an inverse pair) the cost replicates: |.| 0.931, 1/8 vs signed 1.000, 8/8 (-0.069 [-0.095, -0.045], Fisher p 0.0014). With moves +1/+2, which needs integration (index RoPE 0.649, 0/8), |.| scores 0.883, 2/8 against signed 0.947, 5/8 (-0.064 [-0.200, +0.075], Fisher p 0.31): the registered comparison is not resolved, because +1/+2 (every revisit a lap of at least 16 moves) is harder for the signed step as well. Signed models on +1/+2 end with 0.92 of channels having both move angles of one sign (0.01 on +1/-1), the sign-coherent code Lemma 1 allows. The |.| step does not learn +1/+2 more reliably than +1/-1 (2/8 vs 1/8).

H5. Three times the ring budget (`RING_BUDGET_RESULTS.md`, seeds 40-47)

The |.| step at 900 epochs solves 3/8, as at 300 epochs on the same seeds (3/8); the solved sets differ.

H6. Cancelling share and ring strata

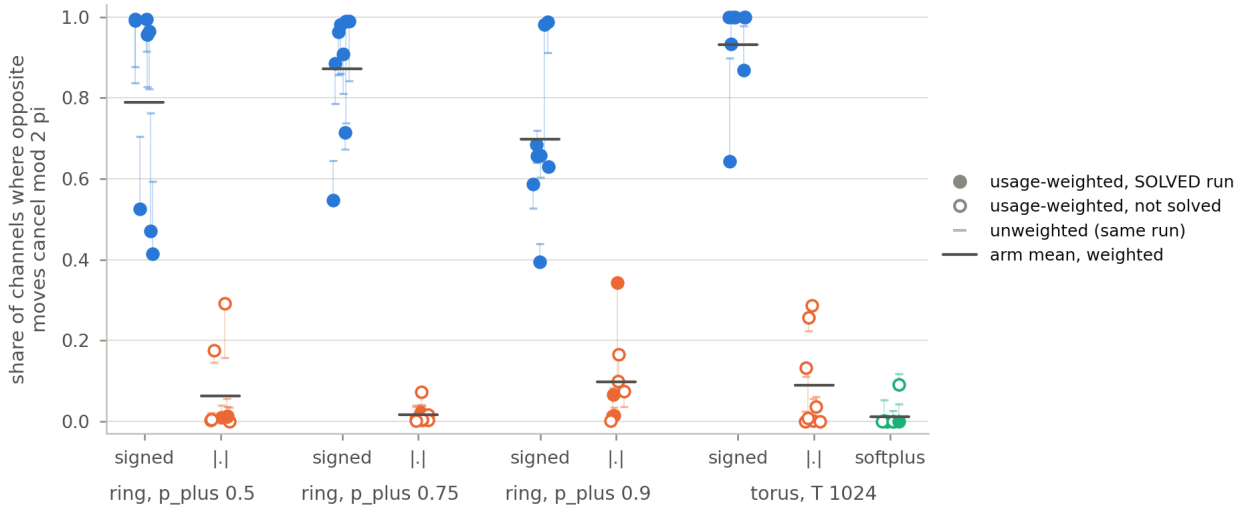


Figure H1 (`fig_wrap_share`; analysis). Share of rotary channels in which opposite moves cancel mod 2 pi, per run, weighted by how much attention uses each channel (dots; filled = SOLVED) and unweighted (faint dashes, joined to the same run). A channel cancels when |wrap(a(move) + a(opposite move))| < 0.1 rad; channels whose move angle has collapsed to within 0.05 rad of 0 mod 2 pi are excluded from the denominator. Usage weight: mean query norm of the channel at move tokens times mean key norm at observation tokens. Ring of 32 cells at three move biases (n = 8 each) and the torus at T = 1024 (n = 8). Bars: arm means of the weighted share. Signed steps 0.70-0.93; |.| steps 0.02-0.10; softplus 0.01. Seven of the eight non-negative runs that solve stay at or below 0.07.

cell	arm	SOLVED	weighted mean (range)	unweighted mean (range)
ring p_plus 0.5	signed	8/8	0.79 (0.41-0.99)	0.79 (0.59-0.92)
ring p_plus 0.5	|.|	2/8	0.06 (0.00-0.29)	0.06 (0.02-0.16)
ring p_plus 0.75	signed	8/8	0.87 (0.55-0.99)	0.78 (0.64-0.86)
ring p_plus 0.75	|.|	1/8	0.02 (0.00-0.07)	0.03 (0.02-0.04)
ring p_plus 0.9	signed	8/8	0.70 (0.39-0.99)	0.64 (0.44-0.91)
ring p_plus 0.9	|.|	4/8	0.10 (0.00-0.34)	0.06 (0.02-0.10)
torus T 1024	signed	8/8	0.93 (0.64-1.00)	0.98 (0.90-1.00)
torus T 1024	|.|	0/8	0.09 (0.00-0.29)	0.11 (0.00-0.27)
torus T 1024	softplus	1/8	0.01 (0.00-0.09)	0.03 (0.00-0.12)

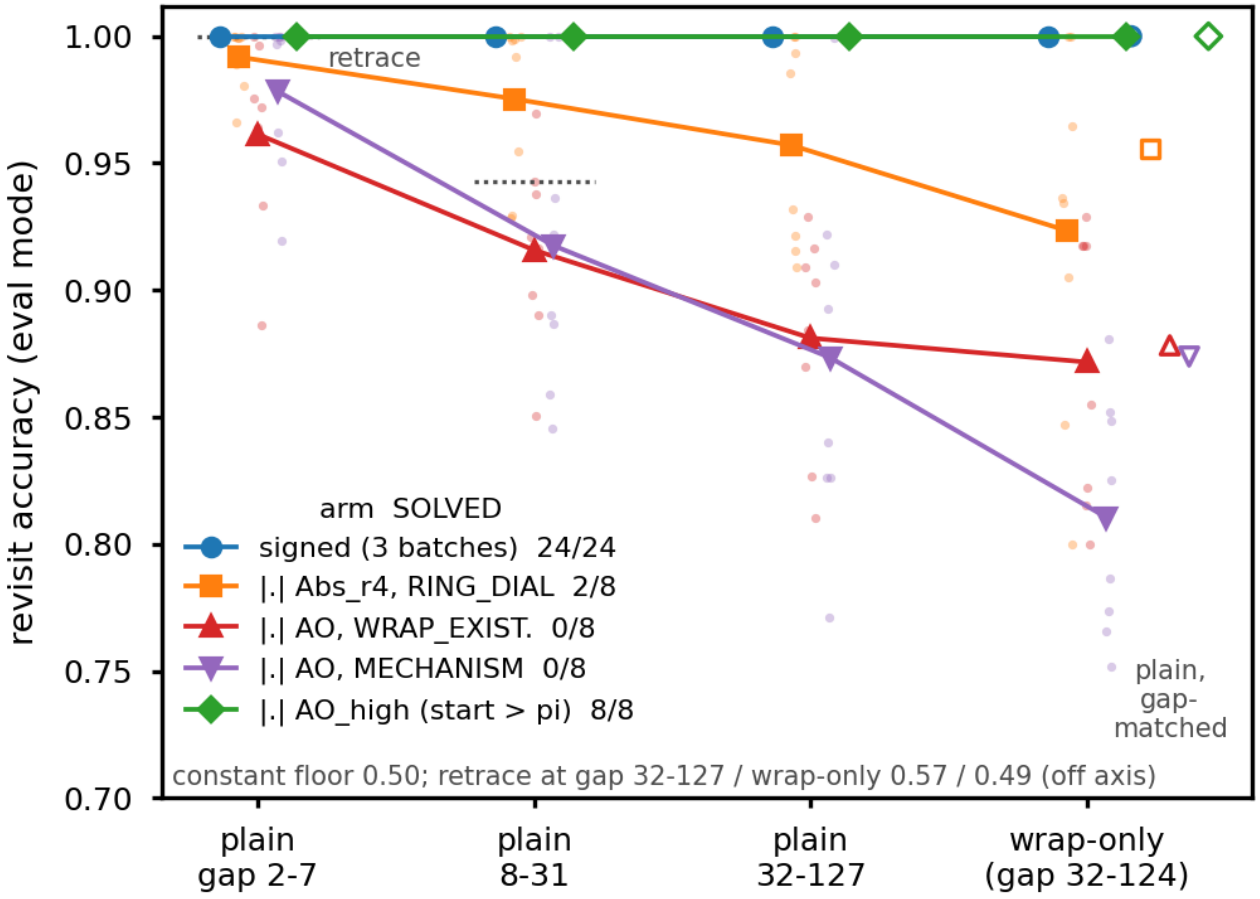


Figure H2 (`fig_ring_strata`; analysis). Revisit accuracy on the 1D ring by stratum (eval mode; 400 trajectories, T = 128; maps redrawn). "Plain" revisits have an earlier visit at the same unwrapped position; "wrap-only" revisits were seen only a lap away. Lines: arm means; dots: runs; hollow markers: plain revisits re-weighted to the wrap-only gap distribution. Signed steps (24 runs over three batches) and |.| steps started past pi are at 1.000 everywhere. |.| steps from the default start lose accuracy with the gap (0.96-0.99 at gaps 2-7, 0.87-0.96 at gaps 32-127). Dotted: retrace within each stratum, 1.000 at gaps 2-7 and 0.943 at gaps 8-31 (0.57 and 0.49 in the long-gap and wrap-only strata, off axis); constant floor 0.50.

I. CARoPE and CoPE

CARoPE (`model_mono_baselines.py`, `CARoPEF`). Per head h and token t a scalar rate f_h(x_t) = 1 / (softplus(x_t W)_h + 1) in (0, 1) replaces RoPE's base frequency, so channel i advances by f_h(x_t)^i per token; initialised as index RoPE (W = 0, a per-head bias with 1 / (softplus(b) + 1) = 10000^(-2 / d_head)). Every token, observations included, advances the phase; the layer is MapWM's, so only the phase differs. Proposition 2 (E5): no channel can read every return. Registered (GENERALITY): torus 0.667 (0/8), below index RoPE (0.722, -0.055, p 0.040); ring 0.741 against index 0.730 (+0.011, not resolved).

CoPE (`CoPEF`). Golovneva et al.'s Listing 1 in torch: per layer one module shared by the heads; gates g_ij = sigmoid(q_i . k_j / sqrt(d_h)); count p_ij = sum_{k=j}^{i} g_ik, clamped to 2047; position logit by linear interpolation of q_i . e[floor p] and q_i . e[ceil p],

table e at zeros initially; added to the attention logits; no rotary position. *Representability (constructed):* gates 17/16 and 15/16 per (move, observation) pair make p_ij mod 2 a function of net displacement, read by a period-2 table (1.000 revisit accuracy). The exact solutions form a lattice (P7 of v3: period 2 or 3).

batch	task	CoPE	SOLVED	comparison
GENERALITY (700-707)	ring	0.887	0/8	signed 1.000, 8/8 (-0.113, p 0.0002); index 0.730 (+0.157, p 0.0002)
WRAP_COPE (900-907)	ring, table at zeros / sinusoid past pi / below pi / slope-matched	0.840 / 0.913 / 0.867 / 0.943	1/8, 2/8, 1/8, 0/8	contrasts not resolved (bimodal seeds)
ACCOUNTS T4 (1100-1115)	256-cell line (never wraps) / ring	0.914 / 0.873	1/16 / 0/16	line: signed 1.000, 8/8 (Fisher p < 1e-4); index 0.772, 0/8

On both line and ring several CoPE seeds reach 0.99-1.00 held-out accuracy without meeting the loss criterion (at a held-out accuracy cut of 0.999 the line reads 5/16 and the ring 2/16; analysis). MapFormer v4 Table 2 reports CoPE failing 1D and 2D navigation (one layer 0.63 +/- 0.21 in 1D and 0.65 in 2D; four layers 0.88 and 0.76).

J. Text world

Task (`TEXTWORLD_RESULTS.md`, `train_textworld.py`). The 64 x 64 torus walk rendered as English: 58 words, three synonyms per direction, movement-free verbs, fillers and asides; T = 1024 words (about 132 steps); held-out map; predict the observation word at revisits. A direction word appears only inside a movement clause (context-free steps). Floors: best constant 0.505, reversal-copy 0.597. Recipe: the torus recipe (A2), path MapWM r4 at one layer against index RoPE at one and two layers; 200 evaluation walks.

Step tables per seed (declared secondaries, `fig_textworld_out.txt`; "minus common" = opposition after removing the component shared by the four direction steps; ablation columns: held-out accuracy, eval mode without the dropout scale):

seed	type	raw opposition	minus common	common / north	accuracy	non-direction steps zeroed	direction steps zeroed
0	clock	1.730	0.0071	0.711	0.881	0.191	0.388
1	map	0.083	0.0029	0.041	0.998	0.893	0.505
2	clock	1.434	0.0064	1.144	0.999	0.665	0.482
3	clock	1.707	0.0022	0.480	0.995	0.673	0.463
4	clock	1.768	0.0052	0.501	0.999	0.539	0.445
5	map	0.120	0.0050	0.060	1.000	0.898	0.505
6	map	0.080	0.0117	0.040	1.000	0.950	0.505
7	map	0.119	0.0057	0.059	0.882	0.586	0.498

On the clock seeds the common component is parallel to the verbs' step (cosine +0.999 to +1.000), and 31-38 of 64 phase channels drift between visits to the same cell (2-4 on the map seeds). Synonym cosine 0.9996-1.0000 on every seed.

K. What and where (analysis)

Do trained path models need the content x position interaction? (`docs/WHAT_WHERE_CHECKS.md`; eval-only on the 48 short-walk torus checkpoints of PAPER2X2.) Held-out accuracy when the score is rebuilt (floors: n-gram 0.598, blank 0.507):

score used	RoPE	PoPE	MapWM r2	MapWM r4	MapPoPE r2	MapPoPE r4
intact	0.805	0.679	0.971	1.000	0.999	1.000
position only (object identity removed, token type kept)	0.463	0.476	0.837	0.989	0.974	0.986
shared position kernel x content gain	0.430	0.435	0.791	1.000	0.988	1.000

Converged path models keep 0.974-0.989 with object identity removed and 0.988-1.000 with a shared kernel times a content gain; index models fall below the floor. Separation as such is MapFormer's (their Fig. 9).

Converging on new objects (`LEAK_RESULTS.md`, `GAIN_PHASE_RESULTS.md`, registered). On a task with a fresh map and 16 fresh objects per sequence (32 x 32 torus, T = 1024, rank 4), object tokens leak a small step into the phase. Removing the step from object tokens (ActOnly) or normalising the step input (NormStep, the step reads LayerNorm(embedding)) lets the model converge: 16/16 SOLVED (unseen-object accuracy 0.9997 / 0.9998) against 0/8 for MapWM (0.9890, still descending). On fresh seeds NormStep again converges 8/8 against MapWM's 0/8 (unseen-object accuracy 0.9940 against 0.9895).

L. PoPE's score

On the short-walk torus (T = 128, rank 2, n = 16, fresh seeds), MapPoPE's gain over MapWM comes with PoPE's score rule (+0.0243, permutation p 0.012; SOLVED 16/16 vs 10/16), and doubling the angle count adds +0.0002 at a ceiling where both arms sit at 0.9995-0.9997 (`MAPPOPE_PAIR_RESULTS.md`).